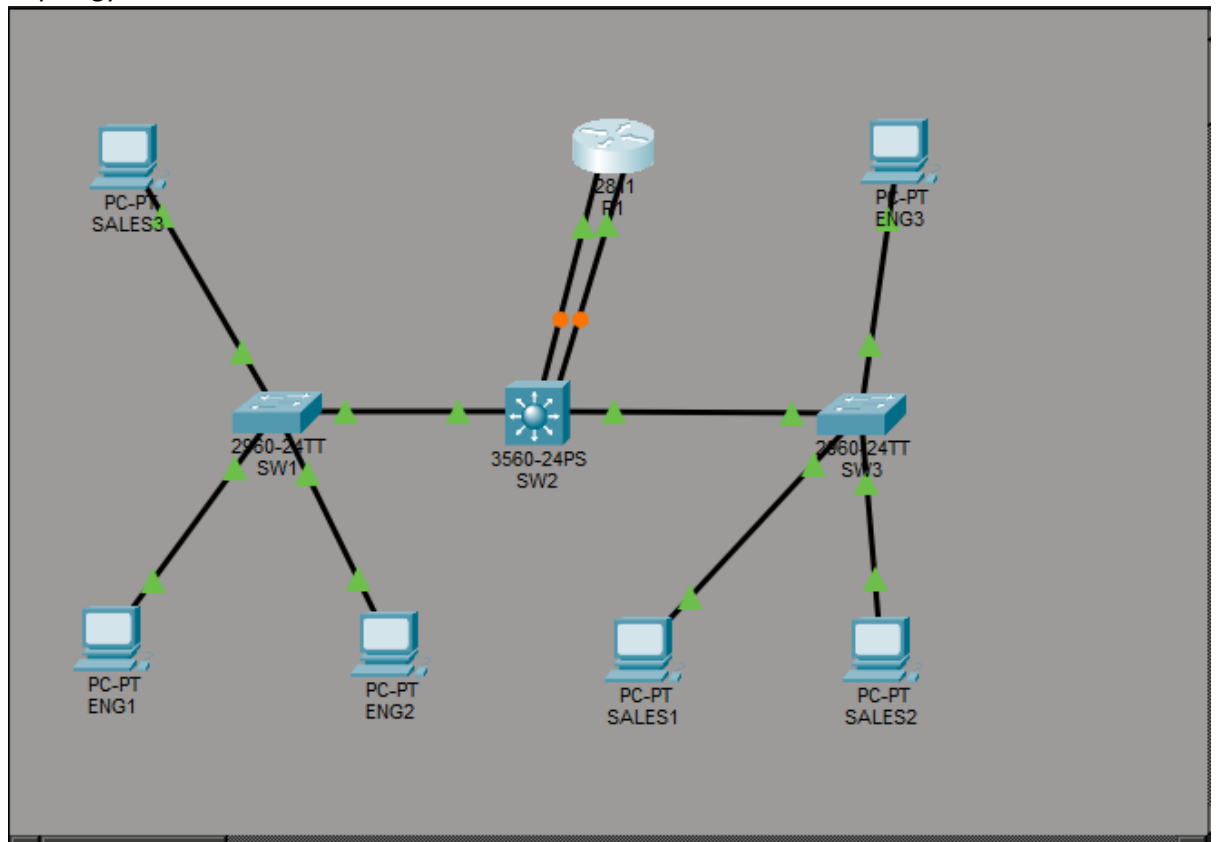


Week 9 practical screenshot and reflection questions:

1.) Topology:



2.) Configuring VTP for the routers:

```
Switch>ENABLE
Switch#CONFIGURE TERMINAL
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vtp domain CAMPUS
Changing VTP domain name from NULL to CAMPUS
Switch(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to CAMPUS.

Switch(config)#vtp mode server
Device mode already VTP SERVER.
Switch(config)#vtp password cisco123
Setting device VLAN database password to cisco123
Switch(config)#exit
Switch#
```

3.) Creating VLANs:

```
Switch>ENABLE
Switch#CONFIGURE TERMINAL
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vtp domain CAMPUS
Changing VTP domain name from NULL to CAMPUS
Switch(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to CAMPUS.

Switch(config)#vtp mode server
Device mode already VTP SERVER.
Switch(config)#vtp password cisco123
Setting device VLAN database password to cisco123
Switch(config)#exit
Switch#
```

4.) Assigning access ports to correct VLANs:

SW1:

```
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface 0/1
^
% Invalid input detected at '^' marker.

Switch(config)#interface0/1
^
% Invalid input detected at '^' marker.

Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
```

SW2:

```
Switch(config)#interface gigabitEthernet0/1
Switch(config-if)#switchport trunk native
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1
(1), with Switch GigabitEthernet0/1 (99).
vl
% Incomplete command.
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gigabitEthernet0/1
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be
configured to "trunk" mode.
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be
configured to "trunk" mode.
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#exit
Switch(config)#
```

5.) Configuring trunk ports: SW1 and SW2

```

Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface gigabitethernet
% Incomplete command.
Switch(config)#interface gigabitEthernet0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#exit
Switch(config)#

```

6.) Verifying VLAN and trunk configuration:

SW1:

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10	MGMT	active	Fa0/1, Fa0/2
20	STAFF	active	
30	STUDENTS	active	Fa0/3
40	GUEST	active	
99	NATIVE	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW3:

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	MGMT	active	Fa0/3
20	STAFF	active	Fa0/1, Fa0/2
30	STUDENTS	active	
40	GUEST	active	
99	NATIVE	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

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7.) Configuring vlan on SW2:

```

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

Switch(config-if)#ip address 10.10.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

Switch(config-if)#ip address 10.10.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 30
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up

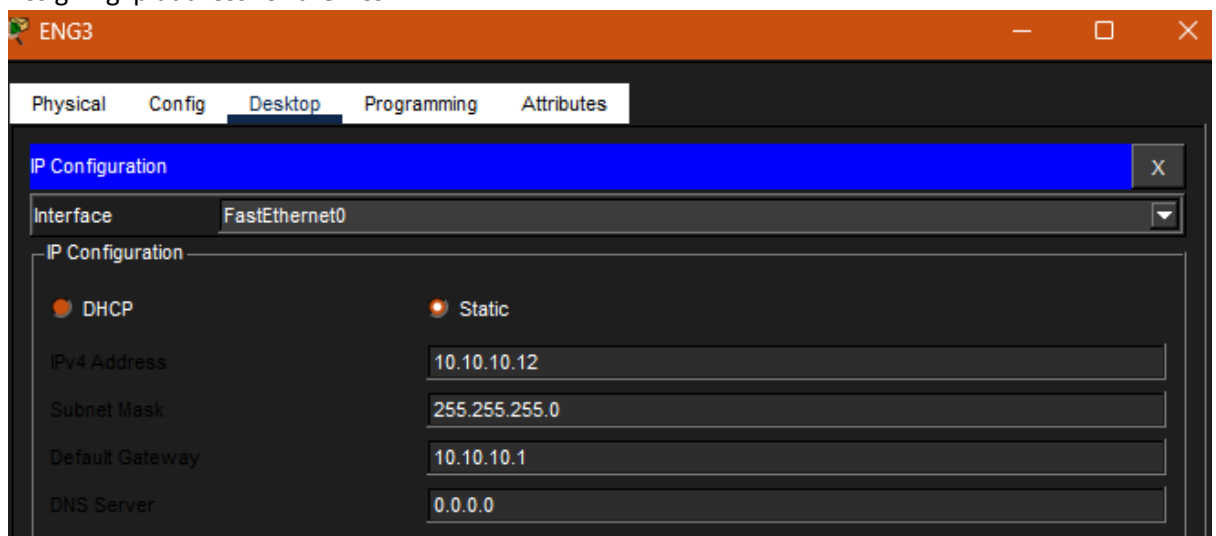
Switch(config-if)#ip address 10.10.30.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 40
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up

Switch(config-if)#ip address 10.10.40.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write memory

```

8.) Assigning ip address for the PCs:



9.) Testing layer 3 switch:

```

Switch>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 4 subnets
C       10.10.10.0 is directly connected, Vlan10
C       10.10.20.0 is directly connected, Vlan20
C       10.10.30.0 is directly connected, Vlan30
C       10.10.40.0 is directly connected, Vlan40

```

10.) Creating sub-interfaces:

```

Router(config)#interface fa0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface fa0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 10.10.10.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface FastEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to
up

Router(config)#interface fa0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 10.10.20.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to
up

Router(config)#interface fa0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 10.10.30.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface FastEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30, changed state to
up

Router(config)#interface fa0/0.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 10.10.40.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface FastEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.40, changed state to
up

Router(config)#interface fa0/0.99
Router(config-subif)#encapsulation dot1Q 99 native
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface FastEthernet0/0.99, changed state to down

```

Question 99: using layer 3 switch is the best option because it reduces cable requirements because routing is performed directly on the switch by using SVI's.